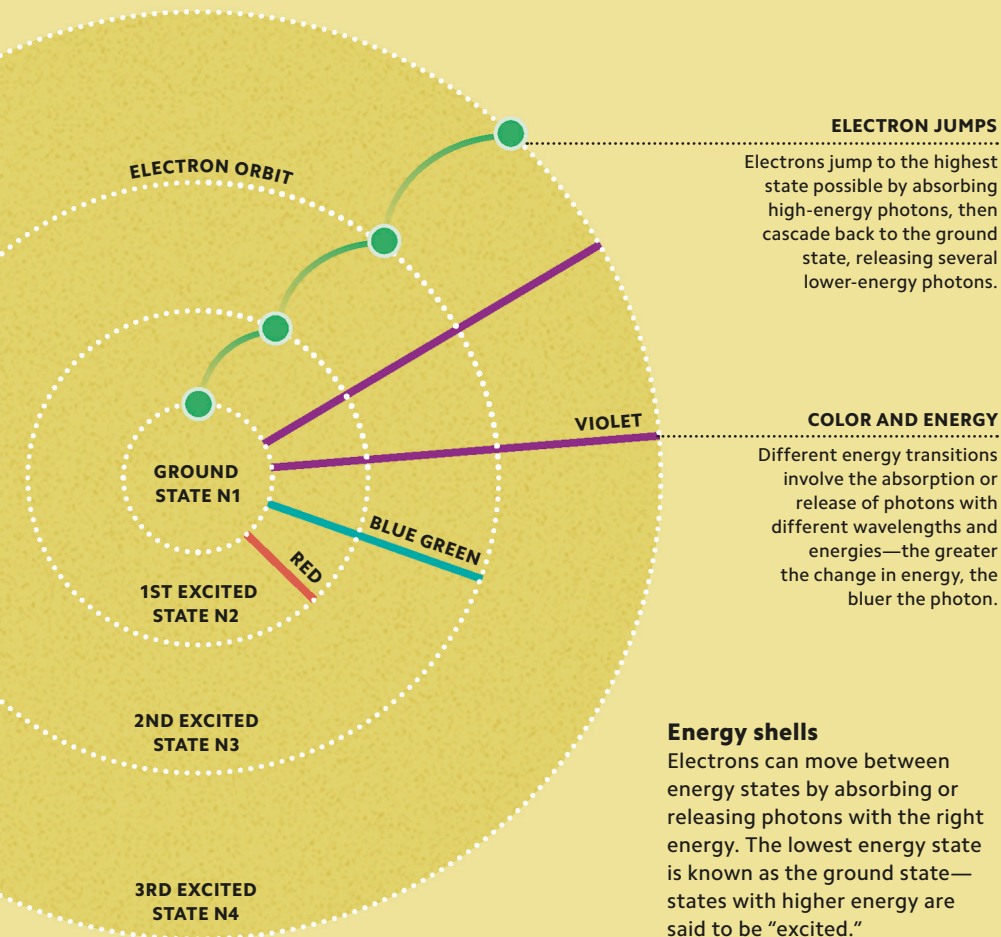




ENERGETIC STATES

Early 20th-century physicists wrestled with how the structure of atoms (see pp.10–11) related to the way they emitted or absorbed radiation. In 1913, Niels Bohr proposed a model in which electrons orbited in shells at various distances from the nucleus, giving each a distinctive energy state. Atoms absorbed or emitted quanta of electromagnetic energy whose wavelengths corresponded to the difference between these states.